Atrazine Degradation System

The degradation of `s-triazine` herbicides, such as `atrazine`, under anaerobic conditions in soil can be described by only two major reaction types (Lit: Erickson LE, Lee KH and Sumner DD (1989), Degradation of atrazine and related s–triazines, Crit Rev Env Control 19:1-14 doi: 10.1080/10643388909388356

For simulate the degradation dynamics using the following rate constants:

```
k_hydrolysis        = 5.00*10^-9
k_de-ethylation     = 3.32*10^-8
k_de-isopropylation = 2.65*10^-8
```

Static Chemical Space Expansion

```
~$ mod -f flow.py
~$ # open the file summary/summary.pdf
```

The derivation graph of the `atrazine` degradation system is generated and all possible 20 degradation pathways from `atrazine` (molecule A) to `cyanuric acid` (molecule P) are calculated via the hyperflow framework.

`summary/summary.pdf` should look like `atrazine-static.pdf`.

Julia ODE Model of the Atrazine Degradation System

```
~$ julia
julia> include("atrazine.jl")
```

A window showing the time course should open after a while. It should like similar to `atrazine-ODE-log.pdf`. Close the window and type `exit()` on the Julia prompt to exit Julia.

Rule-Based Stochastic Simulation (closed system)

```
~$ mod -f stochsim_closed.py
~$ # open the file summary/summary.pdf
```

`summary/summary.pdf` should look like `atrazine-closed.pdf`.

Rule-Based Stochastic Simulation (open system)

```
~$ mod -f stochsim_open.py
~$ # open the file summary/summary.pdf
```

Setting the inflow and outflow rates of **molecule A** to the following rates results in a novel non-equilibrium steady state (NESS), where intermediate species coexist with the output **molecule P**.

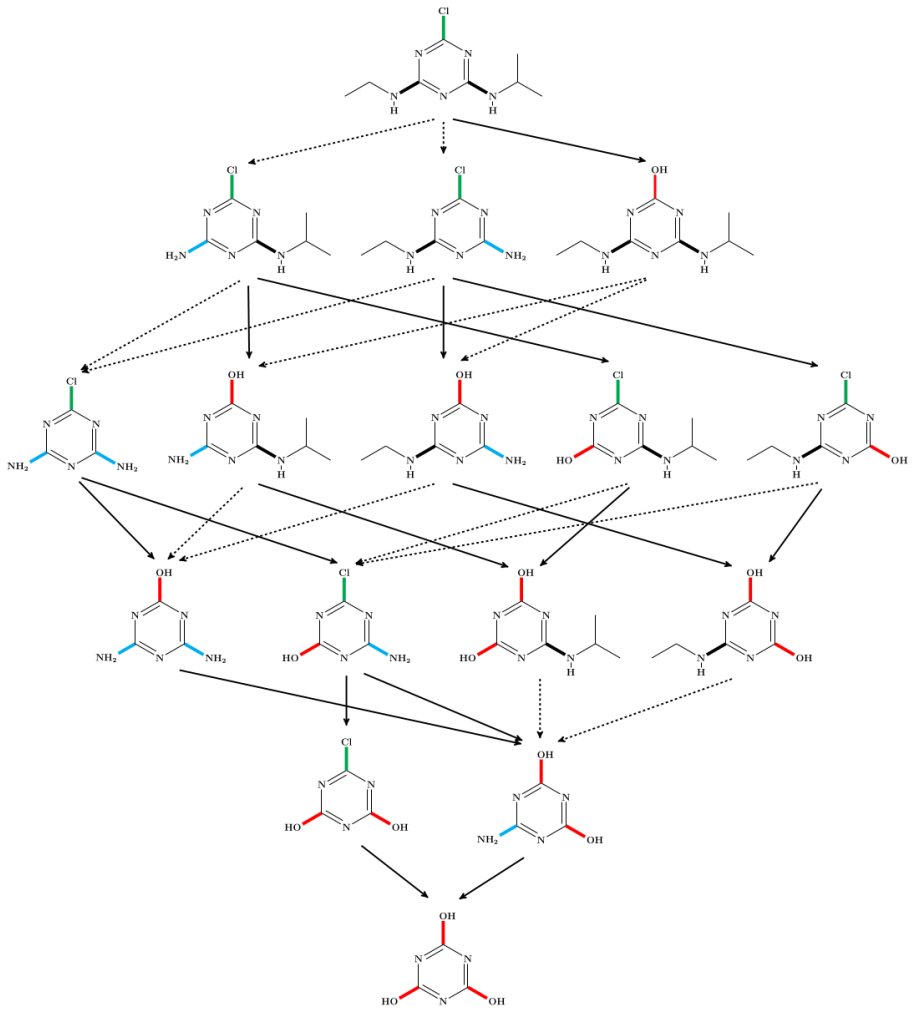

Figure 1: Degradation network compounds are named (A-P) from left to right and top to down. Bold arrows are hydrolysis and dashed arrows reductive de-alcylation reactions.

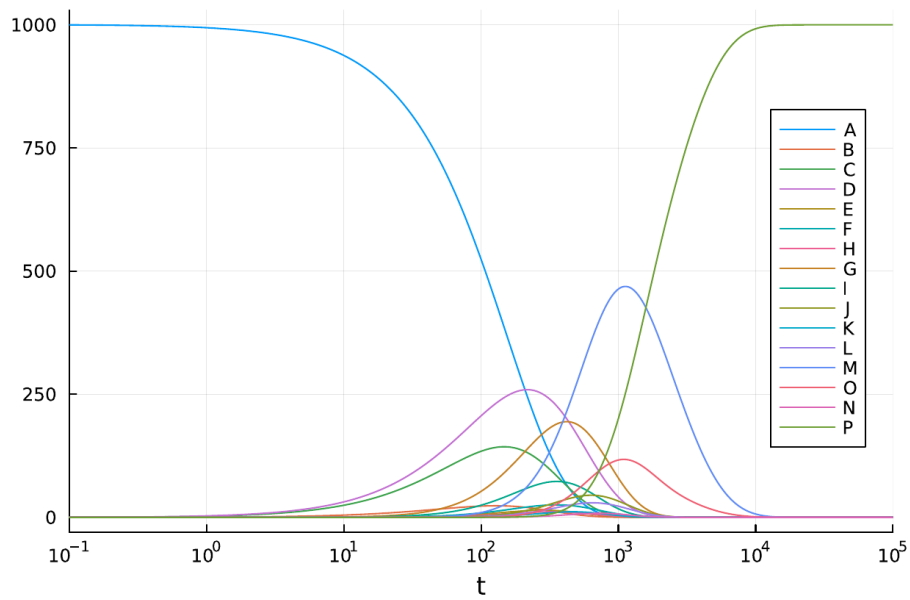

Figure 2: Course of ODE simulation.

```
k_inflow_A  = 4.00*10^-6
k_outflow_P = 1.00*10^-8
```

`summary/summary.pdf` should look similar to `atrazine-open-NESS.pdf`.

Rule-Based Stochastic Simulation (open system on deadlock)

```
~$ mod -f stochsim_open_OnDeadlock.py
~$ # open the file summary/summary.pdf
```

The system starts out as the closed system. When a deadlock is detected, i.e., when no more reactions are possible because **molecule A** is depleted, then the system is opened up, by setting the inflow and outflow rates as in the open system. The system then settles into the NESS.

`summary/summary.pdf` should look similar to `atrazine-open-OnDeadlock-NESS.pdf`.

Further Technical Notes

Use `pdftoppm -png FOO.pdf > FOO.png` to convert a PDF to a PNG. Use `pandoc README.org -o README.pdf` to convert the Org mode file into a PDF

```
=;) xtof
```

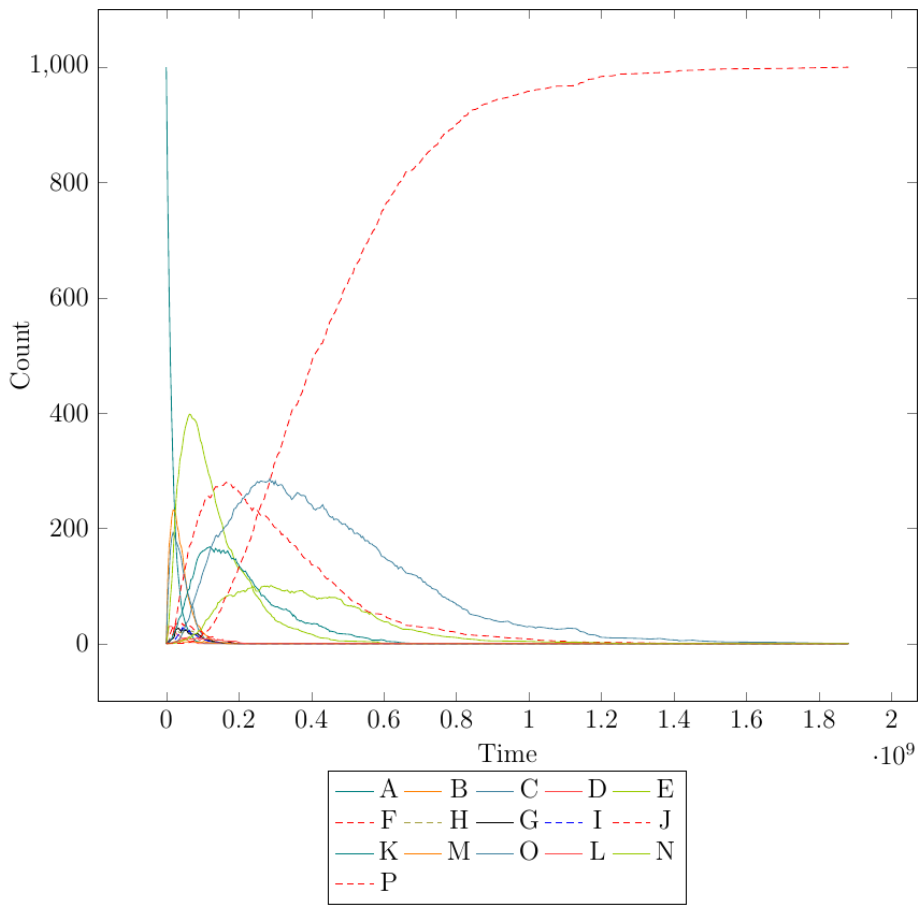

Figure 3: Course of the SS in the closed stochatic simulation.

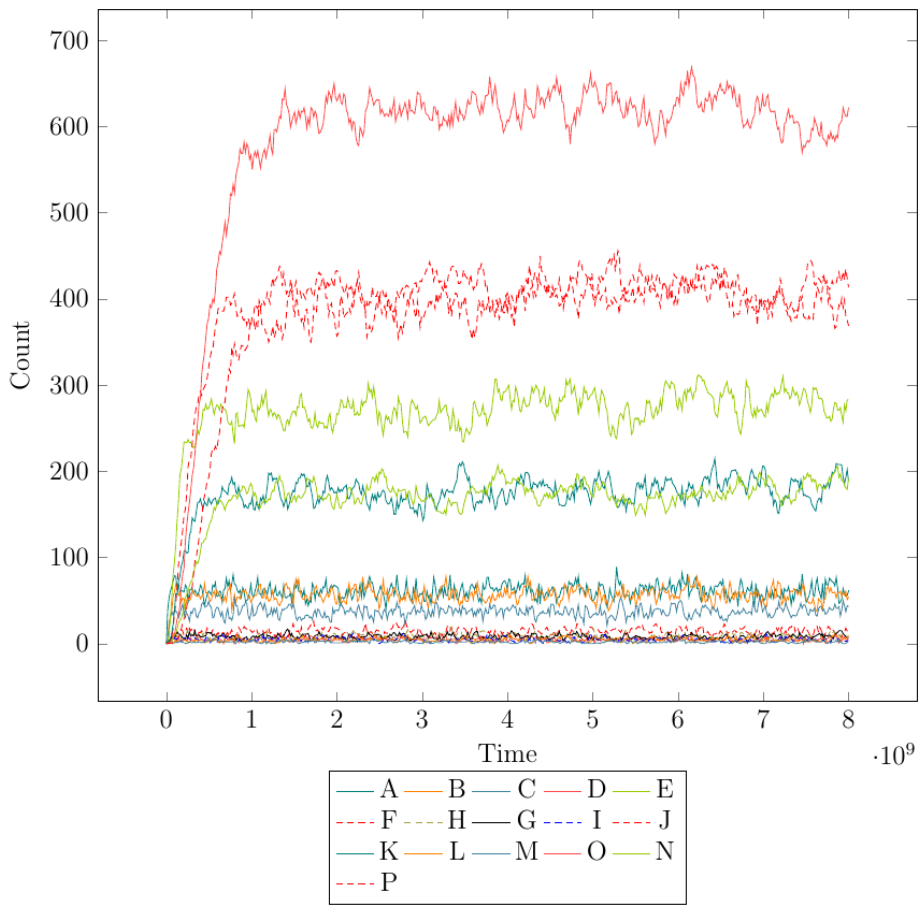

Figure 4: Course of the NESS in the open stochatic simulation.